

Underwriting_Process

Business Process

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内部文档 · 仅供授权人员查阅

Contents

- 1. Overview
- 2. Business Process
 - 2.1 Main Flow
 - 2.2 Detailed Steps
 - 2.3 UW Decision Codes
- 3. System Interaction Standards
 - 3.1 Key Interfaces
 - 3.2 Timeout & Compensation

Underwriting & Risk Assessment Process

1. Overview

This document defines the underwriting business process and system interaction standards, covering the full chain from application entry to policy issuance. It is intended for developers and testers of the underwriting center, risk engine, channel center, and payment center.

2. Business Process

2.1 Main Flow

Application Entry → UW Assessment → Premium Calculation → Payment → Policy Issuance

2.2 Detailed Steps

Step 1: Entry

- Agent/customer enters application via channel system
- Validation: required fields, age/occupation/sum insured rules
- Application number generated, status: Entered

Step 2: UW Assessment

- Auto UW: Rule engine evaluates health declaration, occupation, sum insured
- Approved → proceed to premium calculation
- Declined → notify customer, process ends
- Referred → assign to human UW
- Manual UW: UW reviews health documents, investigation
- Approved/Loaded/Excluded → recalculate premium
- Declined → process ends

Step 3: Premium Calculation

- Calculate premium using product rate table + UW decision
- Adjust rate factors for loaded/excluded cases
- Generate payment notification

Step 4: Payment

- Interface with payment center for first premium collection
- Support: online payment, bank transfer, POS
- Payment success → notify underwriting center
- Payment timeout → application expires

Step 5: Policy Issuance

- Generate formal policy
- Policy number allocation
- Send e-policy to customer
- Notify channel center for commission calculation
- Notify reinsurance center if applicable

2.3 UW Decision Codes

Code	Meaning	Next Step
UW-001	Standard	Normal rate issuance
UW-002	Loaded	Recalculate premium

Code	Meaning	Next Step
UW-003	Excluded	Adjust clauses
UW-004	Deferred	Re-apply when conditions met
UW-005	Declined	Refund premium if paid
UW-006	Referred	Assign human UW

3. System Interaction Standards

3.1 Key Interfaces

Interface	Caller	Responder	Protocol	Description
Create Application	Channel	Underwriting	HTTP/JSON	Create application
UW Request	Underwriting	Risk Engine	gRPC	Initiate auto UW
UW Task Assignment	Risk Engine	Manual UW	MQ	Refer to manual
Premium Quote	Channel	Product	HTTP/JSON	Premium calculation
Payment Request	Underwriting	Payment	HTTP/JSON	Initiate payment
Payment Callback	Payment	Underwriting	MQ	Payment result
Policy Issue	Underwriting	Policy	MQ	Generate policy

3.2 Timeout & Compensation

- UW interface timeout: 5s, convert to async on timeout
- Payment timeout: varies by channel (30s-24h)
- Issue failure compensation: auto-retry 3 times, then manual queue